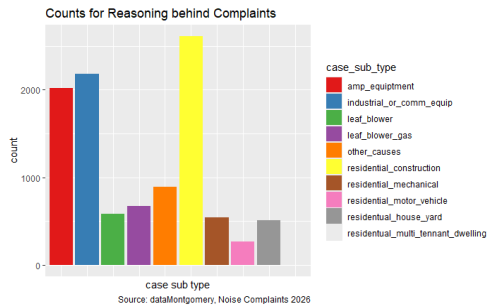


To be noise sensitive “refers to physiological and psychological internal states of an individual that increase the degree of reactivity to noise in general” (Shepherd et al., 2015). One such state can be labeled as misophonia, a relatively new term that has become widely used to describe those with an intolerance to certain sounds—which I have. There are countless other conditions that influence noise sensitivity—such as autism, concussions, Lyme disease, etc—though noise is generally worse for those who are older, since aging is known to cause noise sensitivity with or without hearing loss. Thus, this topic is relevant to a large portion of the population.

Noise has only increased over time, but considering how many are impacted by it, how many people actually know the side effects of that? Noise exposure, especially for those who are noise sensitive, can increase blood pressure, negatively impact sleep, increase heart rate, and even change brain chemistry. It might seem like a small thing, but to be noise sensitive and live near an area with high noise levels can be a miserable experience. So, I would like to give some general advice to those that need it. To do so, I will analyze three dataMontgomery datasets: Noise Complaints, New Home Builders, and Department of Alcohol Beverage Services (ABS) Licensee Data. Noise Complaints contains noise complaints within Montgomery County, MD from 1989 to 2026. New Home Builders contains all of the licenses given to build residential homes in Montgomery County, also from 1989 to 2026. The ABS dataset is a regularly updated list of all the companies within Montgomery County that are licensed to sell alcohol, for example, restaurants, liquor stores, bars, pubs, etc. To perform my analysis on these datasets, I will use R and RStudio.

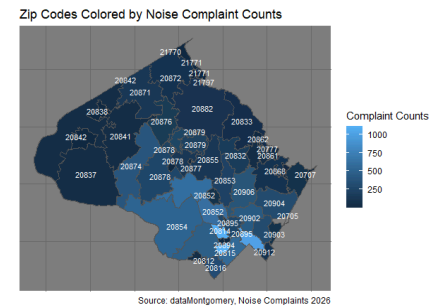
To start off my analysis I cleaned up all of my datasets. To be brief, this entailed handling the NAs, fixing variable names, and selecting relevant columns. I very quickly moved on to

EDA, which is the brunt of what my project is. In doing so I created choropleths, line graphs, and bar graphs—those not displayed here can be found in my GitHub. From the choropleths I found

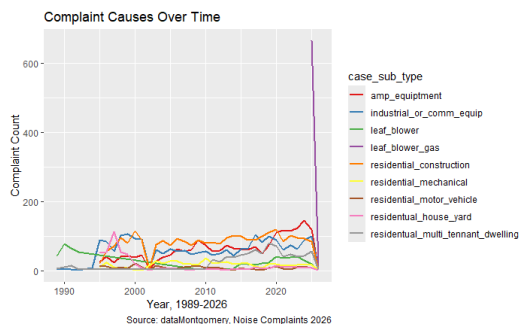
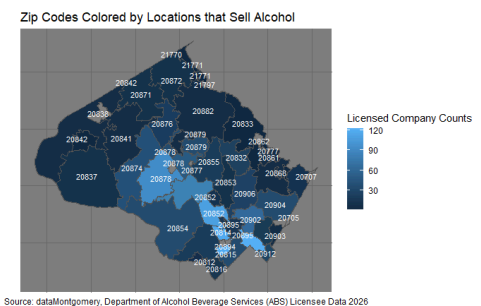
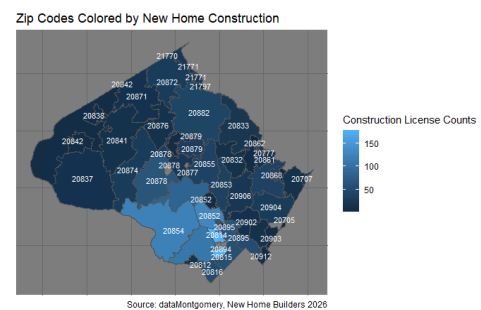


a potential relationship between constriction, locations that sell alcohol, and noise complaints. There is more evidence to support a relationship between construction and noise

complaints than noise complaints and locations that sell alcohol, which can be seen in my bar graph above that shows residential construction is the most complained about overall. Each of the zip codes with higher counts look like a mini version of the above bar graph, so construction is consistently at the top for most complaints.



It's hard to tell just how quickly noise complaints are increasing, so I ran a Negative Binomial Regression model. I ran the model with the counts for the total complaints as the response, and the zip code, year, and complaint causes as predictors. In order to do so, I needed to use 'factor()' on the categorical variables, and I also centered the year to help the model's accuracy, namely preventing



inflating the correlation between interactors. I found that complaints increase by about 2.9 percent yearly, which is less than I thought it would be, but it makes sense when I think about it as a 29 percent increase every ten years. My model also confirmed that Silver Spring (20910) and

Bethesda (20814) are associated with higher counts. My model did output gas powered leaf blowers as null, which can be explained by the line graph to the above left. Since this category only exists in 2025 and 2026, the model cannot properly compute it. I also ran ARIMA to forecast the future, but the forecast ended up flat, which wasn't interesting and only told me that my future will be consistent with my past.

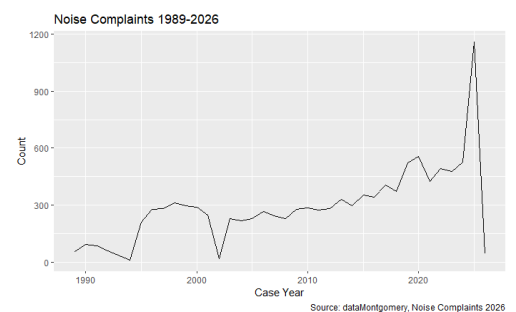
I both looked at a graph from Statistical Atlas of the population density within Montgomery County, and also read several articles that rank the cities within Montgomery County from most populated to least. The population likely has a hand in the distribution of complaints, especially since my dataset was collected via a voluntary report, but it definitely isn't the whole story. There are areas with high population density but almost no complaints, and the population density doesn't overlap nearly as nicely as the choropleth I made from the New Home Builders dataset. From what I've found in my bar graphs and choropleths—the rest of the bar graphs are in my GitHub—it is likely construction and industrial/commercial equipment are the most responsible factors, and are also the most complained about overall.

From my analysis I found that Silver Spring and Bethesda are areas for noise sensitive people to look out for. They have the most complaints, the most construction, and the most locations that sell alcohol—places that likely attract noisy human activity. It is also an intuitive suggestion that noise sensitive individuals avoid big cities, so D.C may be a place to avoid as well, this is backed up by the Noise Complaints choropleth gradually getting more dense towards D.C. Of course, most people cannot pick and choose where they live or where they spend their time, so I'm also choosing to give noise-cancelling headphone recommendations. I recommend Bose QuietComfort headphones for those who wear glasses, as they are known to not hurt the

ears of those who wear glasses. I also recommend Sony WH-1000XM6 headphones, for their performance in both noise canceling and general sound.

For me, the narration within the presentation itself was the hardest part. My past English teachers probably wish they could've held me back based on my narrative writing alone. It took about four drafts of my slides to get to a point where I found it acceptable, and I still have my doubts. However, I do think that it was an overall rewarding experience. I have a stake in the topic, so my interest in it wasn't as forced as past projects, and I actually enjoyed working on it, even when I was struggling with which direction to take it in. Plus, the immense stress from the entirety of this year helped me find new coping mechanisms, which I really needed to do and had been putting off. In the future, I would probably be more ambitious with using more secondary datasets. There are a lot of different complaint causes within the dataset, and I think it would be really cool to look into more datasets to match these causes.

I would recommend that more datasets be taken, which is exactly what I would change if I could do it again. On top of that, I would recommend taking data specifically from the census, which could offer valuable insight on how demographics overlap with complaints. Lastly, I would recommend doing more research into Maryland and Montgomery County history. I couldn't find anything that could explain some of the weird behaviors in complaints during 1994 and 2002—which can be seen in the line graph to the right—but that doesn't mean nothing happened. Government reports are difficult to read all the way through, so it could provide better insight if, in the future, a significant amount of time is taken to slowly look through reports and articles from 1994 to 2002.



I would like to thank all of my past data professors: Mais Alraee, Lori Perine, and Rachel Saidi. They have taught me everything I know about coding and data science—despite my dad working primarily as a data analyst and economic professor, we don't talk about data science. Speaking of my father, I also give thanks to him, Kuo-liang Chang, for telling me that I was doing too much background research and it was taking away from the time I could be spending on the actual coding aspects. My friends also get thanks for their emotional support, specifically talking me down from illogical spirals the whole year. It's been a rough year for all of us, so I really appreciate the amount of time they've spent on me. Of course, I also thank the Montgomery County Government and DataMontgomery for providing me with my dataset and this opportunity. My last thanks is for my early college cohort and the Data 205 class, for all the feedback they've given me and all the fun times we've had these past two years.

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